# Bio-Resonance Platform

Non-invasive developmental metrology built from computationally manufactured resonators, multimodal sensing and temporal AI.

**RESEARCH & PARTNERSHIP DECK**



## ScanIQ LLC

September 2026

01

**IMAGE-LED**

Time-lapse systems infer competence from appearance and division timing.

02

**SPARSE SIGNAL**

Metabolic and genetic assays add information but can be episodic, invasive or expensive.

03

**MISSING LAYER**

Continuous physical-property sensing could reveal deviations before morphology becomes obvious.

Opportunity

## Add a passive, continuous electromechanical phenotype to the embryo-development record.

| 1.49 MHz | 13 stages | 0.92 | ≈510 MPa |
| --- | --- | --- | --- |
| Bovine embryo EIS anchor | Medaka stages modeled | Metabolic + morphology AUC | Fused silk flexural strength |

| | |
| --- | --- |
| **Embryo impedance** | Differential spectra change with cell mass, viability and development. |
| **AI morphokinetics** | Time-lapse models provide independent stage and outcome labels. |
| **Chiral materials** | Fused silk and asymmetric microstructures can rotate THz polarization. |

1

## SDF + material field

Geometry, density, fiber direction and chirality

2

## Field-oriented printing

Aperture deposition with controlled alignment and cure

3

## Differential sensing

Impedance, resonance, phase, Q and near-field channels

4

## Temporal AI

Stage estimate, deviation score and outcome probability

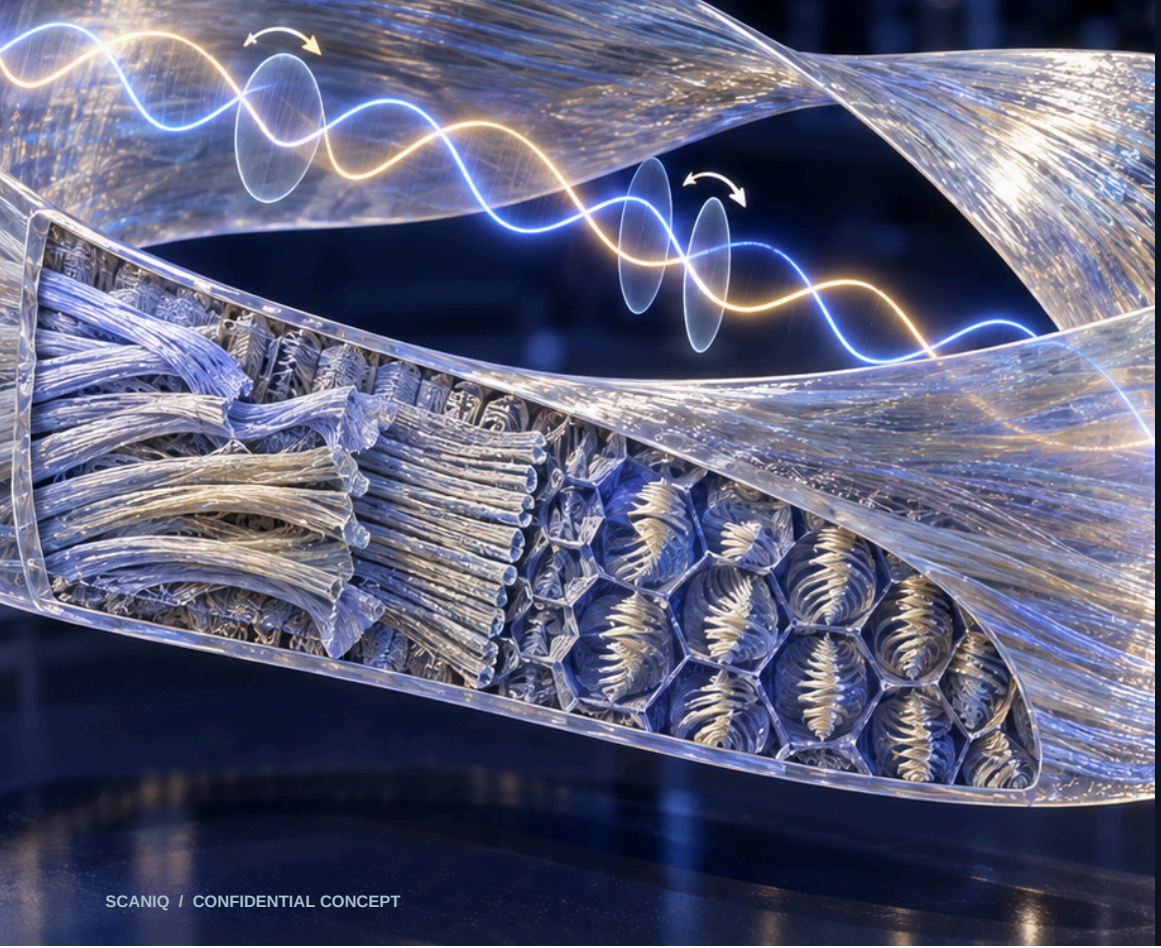
5

## Measured redesign

Update local material orientation toward the target transfer function

**Design intent**

# Maximize developmental-state information per unit exposure, rather than maximize Q alone.



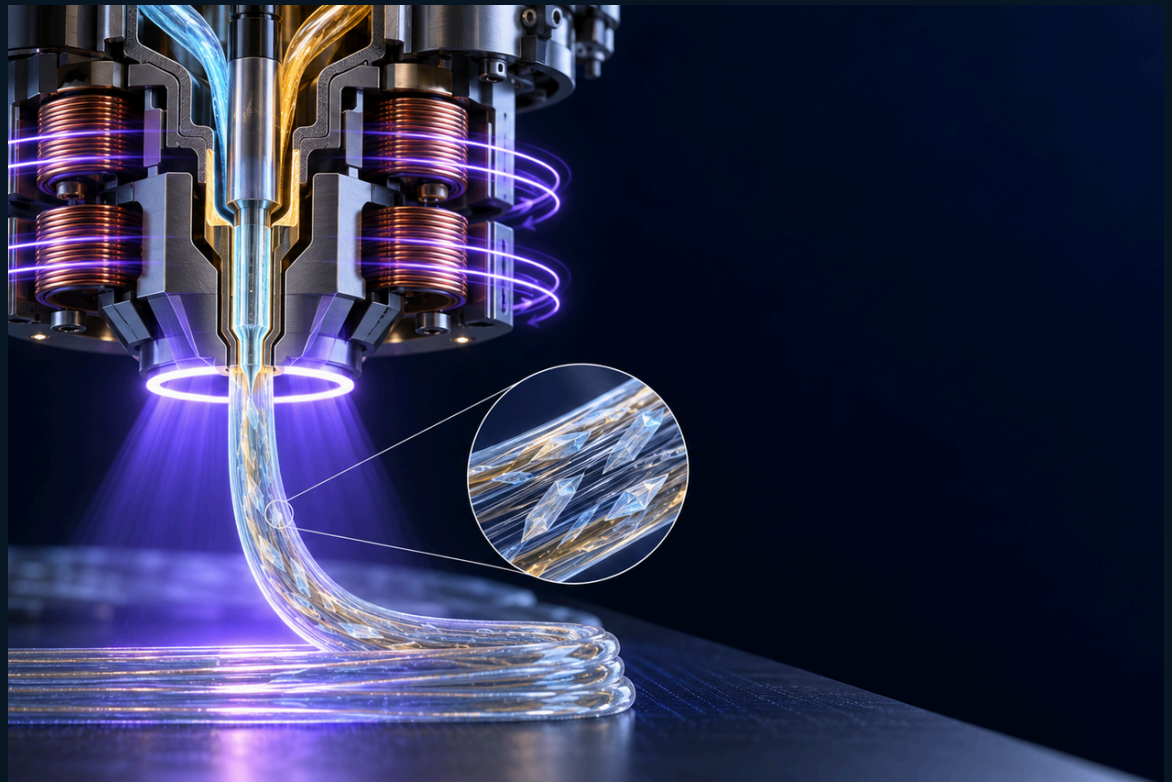
The Möbius form supplies topology. The material field programs local fiber orientation, crystal alignment, handedness and dielectric loss.

**MACRO** Twist, width, thickness, coupling gap

**MESO** Fiber direction, porosity, conductive traces

**MICRO** Chiral domains, crystalline order, additives

**Output: a target electromagnetic transfer function**



**A**

## Feedstock

Silk or polymer composite with alignable fibers or particles

**B**

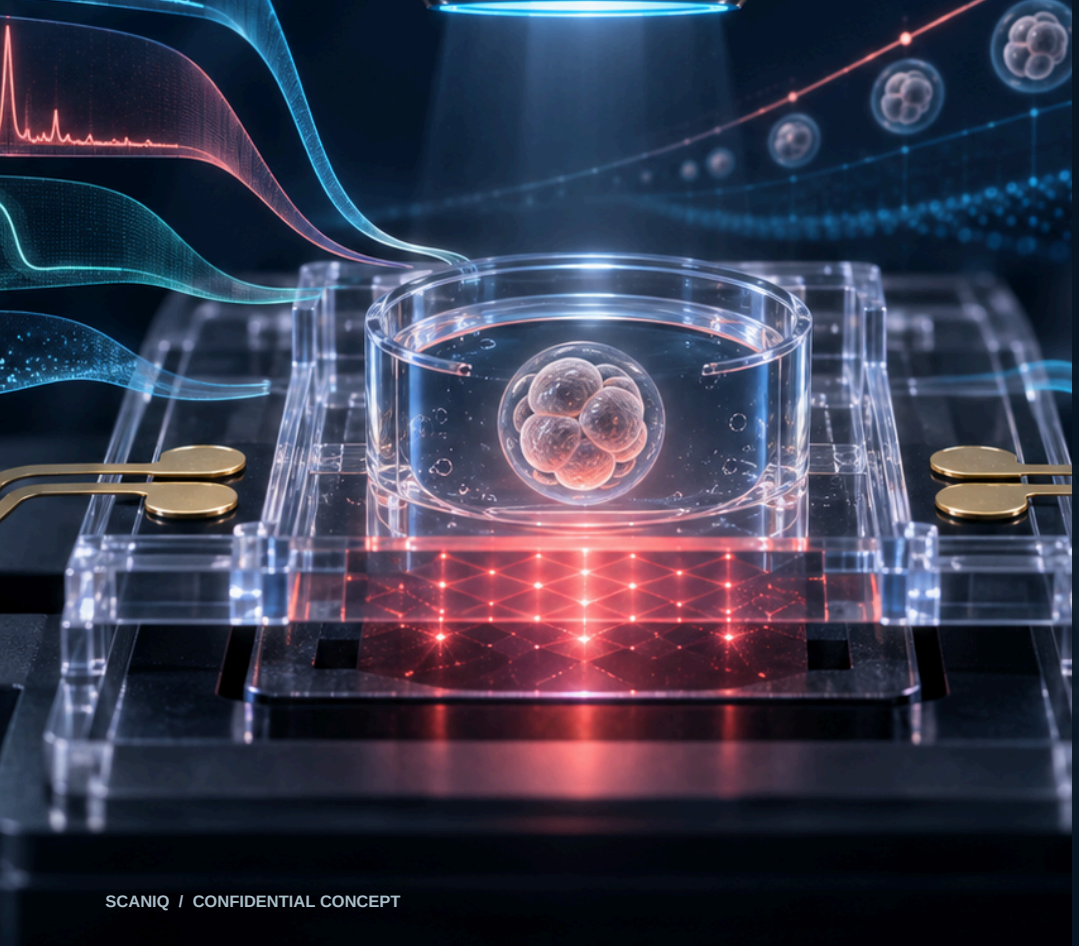
## Orientation zone

Programmed electric or magnetic field follows the voxel vector

**C**

## Fixation

Optical, thermal or chemical cure locks the local structure



### RF / impedance

Δf, ΔQ, S11, phase, complex impedance

### Optical

Morphology, cleavage timing, motion

### Environmental reference

Temperature, vibration, geomagnetic and laboratory EMI

### Optional quantum channel

Near-field NV diamond measurements at realistic stand-off

**THz role**

Interrogate the resonator through near-field or evanescent coupling. Avoid high-loss bulk transmission through culture medium.




MODEL INPUT

**Z(f,t) • $f_r$(t) • Q(t) • phase • image sequence • environment • optional metabolism**

MODEL OUTPUT

**stage probability • trajectory deviation • arrest risk • confidence and uncertainty**

Training principle

**Use blinded outcomes and external validation. Separate discovery models from clinical claims.**

| | | |
|---|---|---|
| 01 | **Bench phantom** | Can the resonator resolve controlled dielectric and geometric perturbations? |
| 02 | **Non-living models** | Do microspheres, cells and organoids produce repeatable spectral signatures? |
| 03 | **Animal embryos** | Do spectra track stage, viability and arrest under blinded conditions? |
| 04 | **Multimodal model** | Does fusion outperform imaging alone on held-out sites? |
| 05 | **Regulated translation** | Can the system meet safety, reproducibility and clinical-study requirements? |

FABRICATION

SDF plus spatial material field controls local anisotropy and chirality

RESONATOR

Topologically chiral hierarchical composite positioned near a biological sample

MEASUREMENT

Differential, broadband, time-resolved perturbation of the resonator-sample system

LEARNING

Temporal multimodal model maps spectra to developmental-state trajectories

REDESIGN

Measured response updates the material field until the target transfer function is reached

A 12-week feasibility sprint can test whether a manufactured chiral resonator adds useful signal beyond standard impedance electrodes.

**MATERIALS**

**Silk/polymer formulation and microstructure characterization**

**FABRICATION**

**Field-assisted printhead, coupon library and metrology**

**BIOLOGY**

**Embryology partner, model selection and blinded endpoints**

**RF / DATA**

**VNA/impedance acquisition, controls and temporal modeling**

**PARTNERSHIP ASK**

**University lab + embryology collaborator + RF/materials fabrication support**